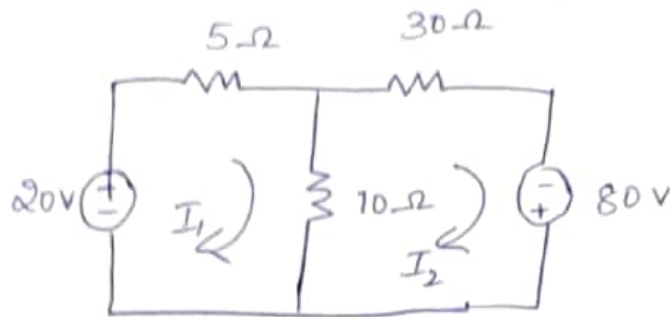


1) Mesh Analysis

(1)

Find the voltage across 30Ω resistor



Loop 1

$$20 - 5I_1 - 10(I_1 - I_2) = 0$$

$$20 - 5I_1 - 10I_1 + 10I_2 = 0$$

$$20 - 15I_1 + 10I_2 = 0$$

$$15I_1 - 10I_2 = 20 \quad \text{--- (1)}$$

Loop 2

$$80 - 10(I_2 - I_1) - 30I_2 = 0$$

$$80 - 10I_2 + 10I_1 - 30I_2 = 0$$

$$10I_1 - 40I_2 = -80 \quad \text{--- (2)}$$

Solving Eqn (1) & (2) we get

$$I_1 = \underline{3.2 \text{ A}}$$

$$I_2 = \underline{2.8 \text{ A}}$$

Current flowing through 30Ω Resistor is I_2

$$\therefore I_{30\Omega} = \underline{2.8 \text{ A}}$$

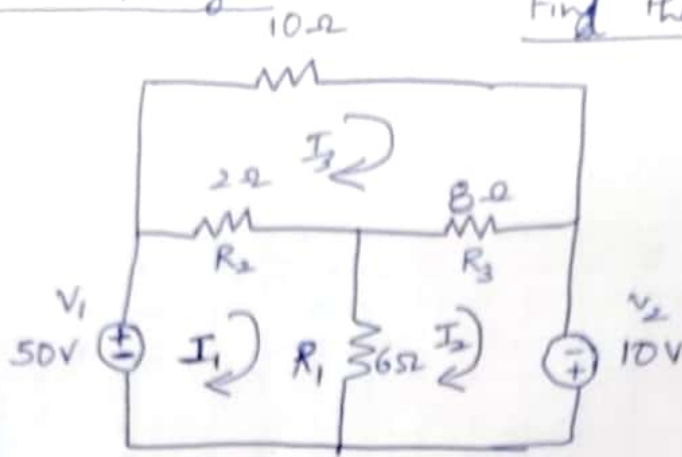
$$V_{30\Omega} = I_2 R = 2.8 \times 30 = \underline{84 \text{ V}}$$

Mesh Analysis

②

Find the loop currents

2)



Loop 1

$$50 - 2(I_1 - I_3) - 6(I_1 - I_2) = 0$$

$$50 - 2I_1 + 2I_3 - 6I_1 + 6I_2 = 0$$

$$50 - 8I_1 + 6I_2 + 2I_3 = 0$$

$$-8I_1 + 6I_2 + 2I_3 = -50$$

$$\boxed{8I_1 - 6I_2 - 2I_3 = 50} \quad \text{--- (1)}$$

Loop 2

$$10 - 6(I_2 - I_1) - 8(I_2 - I_3) = 0$$

$$10 - 6I_2 + 6I_1 - 8I_2 + 8I_3 = 0$$

$$10 - 14I_2 + 6I_1 + 8I_3 = 0$$

$$\boxed{-6I_1 + 14I_2 - 8I_3 = 10} \quad \text{--- (2)}$$

Solving Equations ①, ② & ③

We get

$$\boxed{\begin{array}{l} I_1 = 16 \text{ A} \\ I_2 = 11 \text{ A} \\ I_3 = 6 \text{ A} \end{array}}$$

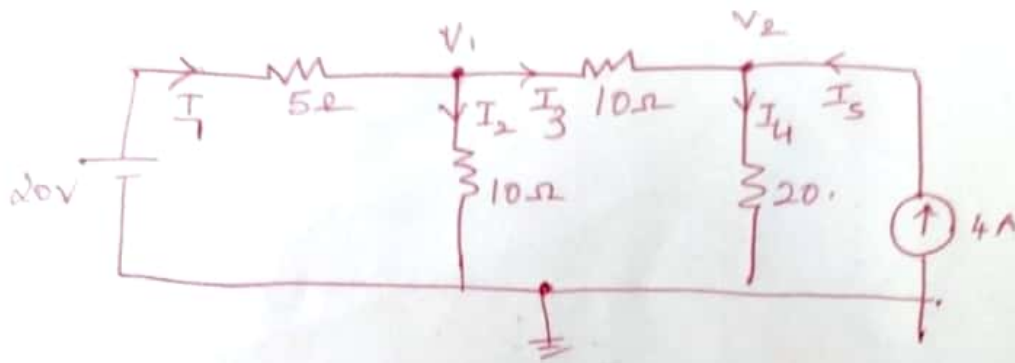
Loop 3

$$-10(I_3) - 8(I_3 - I_2) - 2(I_3 - I_1) = 0$$

$$-10I_3 - 8I_3 + 8I_2 - 2I_3 + 2I_1 = 0$$

$$\boxed{2I_1 + 8I_2 - 20I_3 = 0} \quad \text{--- (3)}$$

① Node Analysis



$$I_1 = I_2 + I_3$$

$$\frac{20 - V_1}{5} = \frac{V_1 - V_2}{10} + \frac{V_1}{10}$$

$$4 - \frac{V_1}{5} = \frac{V_1}{10} - \frac{V_2}{10} + \frac{V_1}{10}$$

$$4 - 0.2V_1 = 0.1V_1 - 0.1V_2 + 0.1V_1$$

$$4 - 0.2V_1 = 0.2V_1 - 0.1V_2$$

$$4 - 0.2V_1 - 0.2V_1 + 0.1V_2 = 0$$

$$-0.4V_1 + 0.1V_2 = -4$$

$$0.4V_1 - 0.1V_2 = 4 \quad \text{--- (1)}$$

$$V_1 = 20V$$

$$V_2 = 40V$$

$$I_3 + I_5 = I_4$$

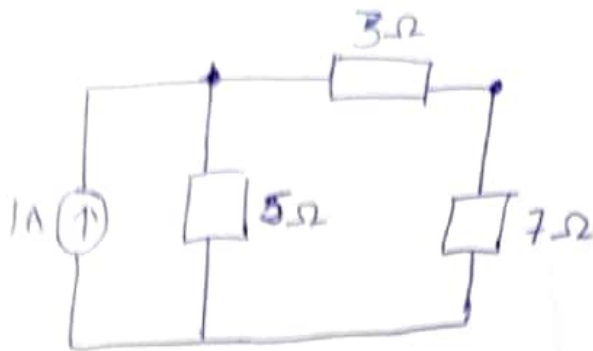
$$\frac{V_1 - V_2}{10} + 4 = \frac{V_2}{20}$$

$$0.1V_1 - 0.1V_2 + 4 = 0.05V_2$$

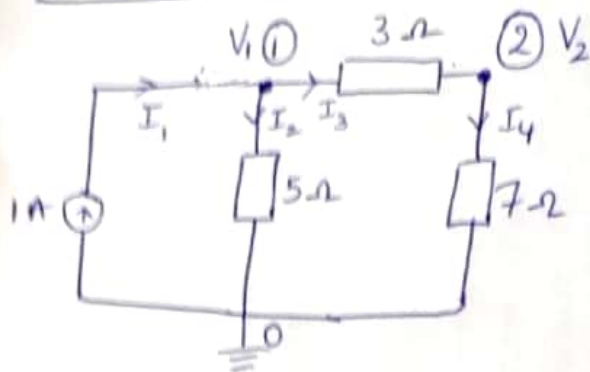
$$0.1V_1 - 0.1V_2 - 0.05V_2 = -4$$

$$0.1V_1 - 0.15V_2 = -4 \quad \text{--- (2)}$$

4)
Node Analysis
Calculate the node voltages.



Solution



Applying KCL at Node 1

$$I_1 - I_2 - I_3 = 0$$

$$I_1 = I_2 + I_3$$

$$1 = \frac{V_1}{5} + \frac{V_1 - V_2}{3}$$

$$1 = 0.2V_1 + 0.333V_1 - 0.333V_2$$

$$\boxed{1 = 0.533V_1 - 0.333V_2} \quad \text{--- (1)}$$

Solving equⁿ ① & ②

$$V_1 = 3.33 \text{ V}$$

$$V_2 = 2.34 \text{ V}$$

Applying KCL at Node 2

$$I_3 - I_4 = 0$$

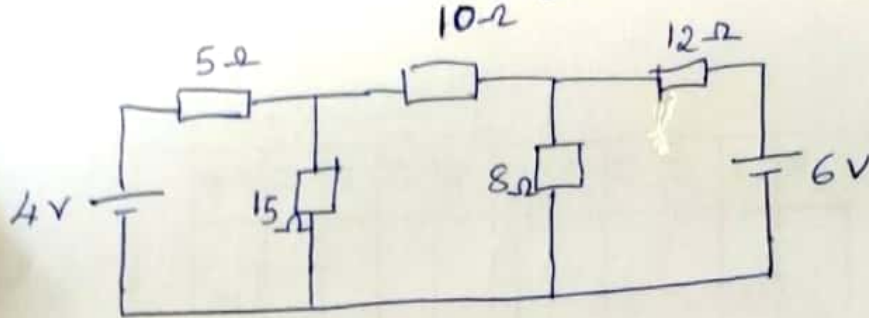
$$I_3 = I_4$$

$$\frac{V_1 - V_2}{3} = \frac{V_2}{7}$$

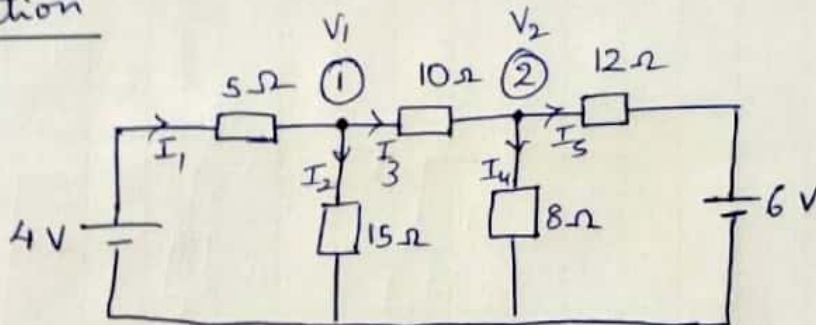
$$0.333V_1 - 0.333V_2 = 0.142V_2$$

$$\boxed{0.333V_1 - 0.475V_2 = 0} \quad \text{--- (2)}$$

Node Analysis : calculate ^⑤ Node voltages & hence the current through 8Ω resistor.



Solution



Applying KCL at Node ①

$$I_1 = I_2 + I_3$$

$$\frac{4 - V_1}{5} = \frac{V_1}{15} + \frac{V_1 - V_2}{10}$$

$$0.8 - 0.2V_1 = 0.066V_1 + 0.1V_1 - 0.1V_2$$

$$0.8 - 0.2V_1 - 0.066V_1 - 0.1V_1 - 0.1V_2 = 0$$

$$\boxed{0.366V_1 - 0.1V_2 = +0.8} \quad \text{--- ①}$$

Solving eqn ① & ② we get

$$\boxed{\begin{matrix} V_1 = 2.88 \text{ V} \\ V_2 = 2.56 \text{ V} \end{matrix}}$$

Applying KCL at node ②

$$I_3 = I_4 + I_5$$

$$\frac{V_1 - V_2}{10} = \frac{V_2}{8} + \frac{V_2 - 6}{12}$$

$$0.1V_1 - 0.1V_2 = 0.125V_2 + 0.083V_2 - 0.5$$

$$\boxed{0.1V_1 - 0.308V_2 = -0.5} \quad \text{--- ②}$$

Current through 8Ω resistor is I_4

$$I_4 = \frac{V_2}{8} = \frac{2.56}{8} = \underline{\underline{0.32 \text{ A}}}$$